
Stage One: Discovery — Completed Workbook

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Every figure in this workbook is produced by `evaluation/discovery_analysis.py` over the 500 development tickets and 29 knowledge base articles, and can be regenerated by running it. Interview quotations are from `05_Datasets/Stakeholder_Interviews.docx`.

Section one: what the interviews tell you

Who	What they told you	What they appear not to know	What you want to verify in the data
Marcus Adeyemi , Head of Support	Over 500 tickets a week against six agents. First response 8–12 hours against a 2-hour agreement. FCR 42% against a 65% benchmark. An escalation costs about four times a resolved ticket. His stated failure condition is a confidently wrong answer: "I would rather it said nothing than said something wrong." Compliance review in the autumn, so decisions must be explainable.	What his tickets are actually about — he says so himself: "If you asked me for a breakdown I would be guessing. We have the data, we have never really sat down with it." He also does not know what is <i>inside</i> his escalations, only that there are many.	The real intent distribution. How much of the escalated volume was avoidable. Whether enterprise customers really are protected.
Sofia Restrepo , Tier 1 agent	40–70 tickets a morning, worked oldest-first. "Seven out of ten I could answer without looking anything up." The time goes on finding and writing the answer, not on knowing it. Keeps a personal snippet file because searching the documentation is painful. Escalates when she does not know, when she is not confident enough to send, or when it is security.	That her seven-out-of-ten estimate is almost exactly right. That her belief about non-fluent customers having the worst satisfaction is not borne out by the record.	The proportion answerable from documentation. Outcomes by language fluency.
Daniel Okonkwo , Tier 2 engineer	"About half of what reaches me is something tier one could have resolved." Escalations arrive as bare forwarded tickets with no summary of what was tried. Wants context, not correctness: "I do not need it to be right. I need it to show its working." Warns that agents' private snippet files contain answers that were correct two years ago.	How large the avoidable share is in numbers rather than impression.	Share of escalations that had a documented answer.
Ines Varga , technical writer	29 reviewed articles covering what support answers repeatedly. Used externally, barely used internally. The cause is search: "Someone writes <i>my deployment keeps dying</i> and my article is called <i>resolving container health check failures</i> . There is no path between those two phrases in a keyword search." Wants to know which article an answer came from. Feature requests, roadmap and novel incidents have no article to retrieve.	That her diagnosis of the mechanism is the central finding of the whole project.	Whether every article is reachable. Which intents have no documented answer.

Who	What they told you	What they appear not to know	What you want to verify in the data
Ravi Menon , customer	Slow but decent. Waiting costs differ enormously by ticket type — a pagination question can wait, a failing deployment at 9am cannot, and they sit in the same queue. Searches the documentation himself and finds the answer about half the time. Will accept an automated first response if it is honest and says where it came from. Wants to know it is automated: "I calibrate how much I trust it." Warns that widening the gap between his business plan and enterprise "will come up at renewal".	That on the historic record enterprise customers are served <i>worse</i> than his own tier, not better.	Resolution rate and time by customer tier.

Where the accounts disagree

The disagreement	Who says what	How the data settles it	What follows from the answer
How much of the volume is routine	Sofia: "seven out of ten I could answer without looking anything up." Marcus: has no breakdown and would be guessing.	71.4% of tickets are labelled answerable from existing documentation. Sofia is right to within a point and a half.	The problem is not a shortage of answers. It is that existing answers are not reaching customers. This reframes the whole project.
Whether escalations are genuinely hard	Daniel: "about half of what reaches me is something tier one could have resolved." Marcus sees only the escalation rate and treats escalations as genuine Tier 2 work.	Of the 281 escalated tickets, 49.11% were answerable from documentation. Daniel's "about half" is accurate; Marcus's implicit assumption is not.	Roughly a quarter of all volume is escalated work that did not need to be. That is the capacity the project can return.
Whether non-fluent customers get worse outcomes	Sofia: "Those tickets have our worst satisfaction scores and I do not think anyone has noticed."	Non-fluent tickets resolve at 45.83% against 43.16% for fluent, with mean CSAT 3.04 against 2.95 and a <i>faster</i> median resolution (198 against 216.5 minutes). Sofia is wrong on the historic record.	Her concern is still worth carrying, but forward not backward: human agents coped by asking questions. A retrieval system that depends on phrasing matching documentation may not, so this becomes a fairness measurement on our own system rather than a finding about theirs.
Whether enterprise customers are protected	Ravi believes his enterprise-plan colleague "gets answers in about an hour". Marcus worries enterprise "will notice immediately if they get worse service".	Enterprise has the worst FCR of any tier at 37.35% (business 48.17%, standard 43.08%) and much the slowest median resolution at 369 minutes against 141 for business.	Both are wrong in the same direction. Enterprise tickets are harder, not better served. Marcus's fear is about a gap that already runs the other way.

What nobody said

What was never mentioned	Why you would have expected it	How you will check it
The documentation-comment channel	It has the worst outcomes of any channel and the longest resolution times. Nobody raised it.	Checked: FCR 34.62% and median 465.5 minutes , both the worst of the four, despite 61.54% being answerable. A ticket raised against the docs is a reader who has already found the page and still needs help.
Repeat contacts	Marcus measures FCR, response time and CSAT. Nobody mentioned customers coming back.	21.6% of tickets are repeat contacts within a week. More than one ticket in five is somebody asking again. Nobody watches this.
Cost or budget for the system itself	Marcus talks about headcount he will not get, but never about what a system may cost to run.	Cannot be answered from the data. Recorded as an open question; the assumption made is that the free-tier constraint in the brief reflects a real cost sensitivity.
Who owns an answer once it is automated	Sofia worries about apologising for something she did not write. Nobody says who is accountable.	Cannot be answered from the data. Recorded as an open question and carried into the governance framework as a named owner per risk.

Section two: what the ticket data shows

What to measure	Your figure	Where it came from	What surprised you about it
Total tickets in the sample	500	<code>development_tickets.json</code>	—
Split by channel	email 212 (42.4%), chat 155 (31.0%), docs_comment 78 (15.6%), forum 55 (11.0%)	<code>channel</code>	Documentation comments are a sixth of volume. Nobody interviewed mentioned them once.
Split by intent	22 intents, none above 5.8%. Largest: data_export 5.8%, data_residency 5.8%, rollback_request 5.6%, deployment_failure 5.4%	<code>labels.intent</code>	A long flat tail, not a few dominant categories. No "top three intents" strategy would cover much of the queue.
Split by urgency	See <code>discovery_findings.json</code> <code>distribution.by_urgency</code>	<code>labels.urgency</code>	—
	43.8%	<code>history.first_contact_resolution</code>	

What to measure	Your figure	Where it came from	What surprised you about it
Proportion resolved on first contact			Close to the 42% Marcus quotes, which is a useful sanity check on the corpus.
Average satisfaction rating	2.97 / 5	history.csat_rating	Lower than the 3.2 Marcus reported. He may be quoting a stale or more flattering figure.
Most frequent single question	No single dominant question; the distribution is flat	labels.intent	—
Proportion answerable from existing documentation	71.4%	labels.answerable_from_docs	The central finding. The answers largely exist already.
Proportion non-fluent, and their outcomes	120 of 500 (24%). FCR 45.83%, CSAT 3.04, median 198 min	language_fluency + history	They <i>do better</i> than fluent customers on every measure. This contradicts Sofia directly.
Outcomes by customer tier	standard 253 (FCR 43.08%, 266 min), business 164 (48.17%, 141 min), enterprise 83 (37.35%, 369 min)	customer_tier + history	Enterprise is worst served, not best.
Proportion that are repeat contacts	21.6%	history.repeat_contact	Not measured by CloudServe at all.
Escalations that had a documented answer	49.11%	cross of history.escalated and labels.answerable_from_docs	Confirms Daniel's estimate almost exactly.
Documentation coverage	All 29 articles are named as an expected source by at least one ticket; no expected id is missing from the corpus	expected_doc_ids against documentation.json	The corpus is complete and internally consistent. No dead articles, no dangling references.

The two rows that matter most

71.4% answerable against 43.8% resolved is a 27.6 point delivery gap. CloudServe do not have an answer problem. They have a problem getting answers that already exist in front of customers. This is the sentence the rest of the project is built on, and it is the difference between building a chatbot that invents answers and building a retrieval system that delivers existing ones.

The non-fluent segment contradicted the person who raised it. Sofia believed those customers had the worst outcomes and that nobody had noticed. The record says the opposite. The value of checking is that it changes what we build: had we taken her account on trust we would have built a special path that the evidence does not justify, and we would have missed the tier gap, which is real.

Section three: where the time actually goes

Effort is approximated by total recorded resolution minutes per intent, the only effort signal the corpus carries. Volume and effort diverge sharply.

Ticket category	Share of volume	Share of effort	Why the two differ	Evidence
security_incident	5.2%	9.69%	786 min mean. Every one needs a human and half have no documented answer.	effort_by_intent
compliance_request	5.2%	8.38%	679 min mean. Contractual and legal review.	effort_by_intent
data_residency	5.8%	7.08%	515 min mean. Daniel: "we get those wrong occasionally even as humans."	effort_by_intent
feature_request	4.0%	6.87%	724 min mean and 0% answerable — there is no article, so time is spent discovering there is no answer.	effort_by_intent
deployment_failure	5.4%	6.18%	483 min mean, but 70.4% answerable.	effort_by_intent
rate_limit	2.6%	0.75%	122 min mean, 92.3% answerable. Cheap and documented.	effort_by_intent
onboarding	4.4%	2.11%	202 min mean, 90.9% answerable.	effort_by_intent
sso_configuration	5.2%	2.36%	191 min mean, 84.6% answerable.	effort_by_intent

What this changes. The four most expensive intents are 20.2% of volume but **32.0% of effort** — and three of the four are ones the corpus says must never be automated. Automation cannot touch the expensive work. What it can do is clear the documented, cheap, high-volume tail so that the expensive work gets the attention it needs. That reframes the business case from "replace agent time" to "return agent time to the work that requires a person".

The steps an agent takes on a typical ticket

Reconstructed from Sofia's account in interview two.

Step	What the agent does	Roughly how long	Could this be automated?
1		seconds	

Step	What the agent does	Roughly how long	Could this be automated?
	Open the queue, sort by age, pick the ticket closest to breaching		Yes, and it should be sorted by urgency rather than age — Ravi's point that a failing deployment and a pagination question sit in the same queue.
2	Read the ticket and work out what is actually being asked	1–3 min, longer where phrasing is unclear	Partly. This is what intent classification does, and it is where the system is strongest.
3	Decide whether she knows the answer confidently enough to send it	under a minute	This is precisely the routing decision, and Sofia's own three criteria — do not know, not confident enough, security — map onto the three escalation rules.
4	Find the answer: search the docs, or more often her personal snippet file	the bulk of the time	Yes, and this is the target. Sofia does not need the answer supplied; she needs it found. Automating this is the whole project.
5	Write it out in her own words	2–5 min	Yes, where the source is a reviewed article.
6	Send, or escalate with no context attached	seconds	The escalation should carry the analysis. Daniel loses time re-reading threads and re-asking customers questions they have already answered.

Steps 4 and 5 are where the minutes accumulate and both are automatable. Step 3 is the one that must remain conservative.

Section four: what the client counts as success

Measure	Who watches it	Current value	What they would call success	How confident are you?
Time to first reply	Executive team; it is in the service agreement	8–12 hours	Under 2 hours contractually; the brief targets under 5 minutes	High — stated by Marcus and consistent with a 214-minute median resolution
First contact resolution	Marcus, and it is what he actually cares about	42% reported, 43.8% measured	60–65%	High — two independent sources agree
Customer satisfaction	Renewal conversations	3.2 reported, 2.97 measured	4.0+	Medium. The reported and measured figures disagree by a quarter point and I cannot tell which is current.
Escalation rate	Marcus, as a cost proxy at 4×	58% reported, 56.2% measured	30% or lower	High
Repeat contacts	Nobody	21.6%	Halved	Medium — measured reliably, but nobody at CloudServe watches it, so there is no target to confirm
Agent attrition	Marcus	Rising; two experienced agents left citing workload	Stable	Low — no data in the corpus, interview only

The sentence test

Prompt	Your answer
This project will have been worth doing if, within three months of launch...	...first contact resolution has risen from 43.8% to above 60%, the escalation rate has fallen from 56.2% below 30%, and the share of escalations that turn out to have had a documented answer has fallen well below the 49.11% it stands at now.
We will know it did not work if we see...	...repeat contacts rising above the present 21.6%, which would mean tickets are being closed without being solved; or any customer receiving an automated answer on a security, compliance or data-residency matter; or the resolution gap between customer tiers widening beyond where it already sits.
The measure the client will actually be judged on internally is...	...time to first reply, because it is the number in the service agreement and the one reported upward. Marcus is explicit that first contact resolution matters more to him, but response time is what is escalated to the executive team. Both must move; only one is contractual.

Section five: data, constraints and risk

Data source	What it contains	Quality and gaps	Access constraints	Private data present?
Support tickets	500 dev / 80 validation, with intent, urgency, expected route, answerability, expected sources, and the historic outcome	Fully labelled and internally consistent. Synthetic, and the intents are unusually separable — 99.8% out-of-fold accuracy will not survive real traffic	Development set free to use; validation for checking only; the hidden set is used once	Customer names, ids, region and tier. Names appear in ticket bodies
Documentation	29 reviewed articles	Complete: every article is an expected source somewhere and no expected id is missing. Ines reviews on a rotation, <code>last_reviewed_days_ago</code> is present	Free to use	None
Past resolutions	200 senior-agent reference answers with required points and forbidden claims	High quality by construction	Free to use, but not a test set	Possible customer references in answer text
Agents' private snippet files	Not supplied	Does not exist as data. Daniel warns it contains answers correct two years ago and wrong since; Ines has twice asked for consolidation	Not available	Unknown
Customer records	Not supplied; tier, region and fluency only, on the ticket	Sufficient for the fairness audit	On the ticket	Yes — tier and region are commercially sensitive

The constraint that shaped the design most. Daniel's warning about the snippet files is the reason retrieval is restricted to Ines's 29 reviewed articles and never learns from historic agent responses. Training on what

agents did would scale up the stale answers he describes, and it would do so invisibly. The cost is that anything not yet documented cannot be answered — accepted, because that case escalates rather than being guessed at.

Initial risk register

What could go wrong	How likely	How bad	Who it affects	First thought on preventing it
The system answers confidently and incorrectly	Medium	Severe — Marcus's stated failure condition; his customers are engineers who "will screenshot a confidently incorrect answer and put it on the internet within the hour"	Customer, then the support team handling the complaint	Calibrated threshold; answers grounded only in retrieved passages; citations checked against what was actually retrieved; block rather than send on any doubt
Private information appears in a reply	Low	Severe — no acceptable rate	Both customers involved; CloudServe legally	Pattern checks on every outbound response, blocking rather than redacting
Some customers get consistently worse answers	High	Severe — Ravi warns it "will come up at renewal"	Whichever segment loses	Segment the evaluation from the start; measure by tier, fluency and channel on every run
Documentation goes out of date	Medium	Moderate	Customers receiving a correct citation to a stale article	Retrieve only from the reviewed corpus; log <code>doc_id</code> on every answer so a bad article can be traced to every reply that used it
Customer text treated as an instruction	Medium	Severe	CloudServe's reputation	Ticket content fenced and passed as data, never as instruction; separate input validation that blocks
A ticket that should never be automated is automated	Medium	Severe	Customer in a security or compliance situation	Hard policy rules on intent, applied before any confidence is consulted, plus a veto when a never-automate intent is a close runner-up
Model provider unavailable	High	Low if handled	Nobody, if it degrades properly	Local grounded generation as a supported mode, not an emergency path

Section six: the problem statement

Element	Statement	Supported by
What the client asked for	A chatbot to answer the easy tickets so agents can do the hard ones	Marcus, interview one
What the evidence suggests they actually need	A retrieval and routing layer that delivers answers which already exist in their documentation, and hands everything else to a person with the analysis attached	§2: 71.4% answerable; §1: Ines on why search fails; §1: Daniel on context-free escalations

Element	Statement	Supported by
The gap between those two	A chatbot generates answers. CloudServe do not lack answers — they lack delivery. A system that generates would invent answers to questions already answered, and would be confidently wrong on the 28.6% that have no documented answer	§2: 71.4% answerable against 43.8% resolved
Who is affected and how	Customers wait 8–12 hours for answers half of them could have found. Agents spend most of each ticket searching rather than answering. Tier 2 spends half its time on avoidable work. Standard and enterprise tiers fare worst	§1, §2, §3
What will change if this is solved	FCR from 43.8% toward 60%+. Escalations carrying analysis rather than arriving bare. Agent time returned to the 20% of volume consuming 32% of effort	§3
What is explicitly not in scope	Security incidents, compliance requests, feature requests and unclear tickets are never automated. No commitments about money, quotas or timelines. No learning from agents' private snippet files. No multi-turn conversation	§1 Daniel; §2 <code>must_not_auto_respond</code> at 100% for all four

The one paragraph version

CloudServe's support team is not short of answers — it is short of a way to find them. Seven in ten of the tickets arriving each week are already answered somewhere in the twenty-nine support articles the company maintains, but fewer than half get resolved at first contact, because the only way to reach those articles is to already know which one you want. So agents rebuild answers from memory and from private notes, customers wait eight to twelve hours for something that was written down two years ago, and half of everything passed to a senior engineer turns out to be work that need never have been passed to anyone. The fix is not a machine that invents answers; it is a way of putting the right existing answer in front of the customer within seconds, being honest when there isn't one, and handing the genuinely difficult cases to a person with the thinking already done.

Testing it against the five

- **Marcus** would recognise it and would be surprised by the seven-in-ten figure, since he said he would be guessing at the breakdown.
- **Sofia** would recognise it immediately — it is her account, quantified.
- **Daniel** would recognise the half-of-escalations claim as his own.
- **Ines** would recognise it as the thing she has been unable to get taken seriously.
- **Ravi** would recognise the waiting, though the framing is about the queue rather than his experience of it.

The statement is not one any of the five could have produced alone: Sofia knows the volume is routine but not why the articles go unused, Ines knows why but not the scale, Daniel sees the consequence but only at his end, and Marcus sees the totals but nothing inside them.

Open questions carried forward

1. Which CSAT figure is current, 3.2 or the measured 2.97?

2. Who is accountable for an automated answer that turns out to be wrong? Assumption: the Head of Support, pending confirmation.
3. What does a wrong automated answer actually cost? Marcus quantified escalations at 4× but not this. The threshold is sensitive to it, so it is carried as a parameter with published sensitivity.
4. Are the agents' private snippet files recoverable for review? If so, the answerable share may be higher than 71.4%.